The empirical recurrence for OEIS A300608, recovered from its tail

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Abstract

OEIS A300608 records "Empirical recurrence of order 72" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 96$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 74$. Its last coefficient is $c_{72} = -1 \neq 0$, so no further tail satisfies anything shorter; any order-72 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A300608 is "Number of n X 4 0..1 arrays with every element equal to 1, 2, 3 or 5 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.". It has offset 1 and begins

$$2, 30, 277, 2347, 19532, 164619, 1387896, 11697909, \ \dots$$

The entry states:

> Empirical recurrence of order 72 (see link above).

As of the "Last modified" line on the live entry (May 25 2024 20:48:02, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 96$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 96$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 72: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 72.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 192$ exact terms from $n = 2$ on, it returns order 72 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{72} c_i\, a(n-i),$$

c_1	9	c_{19}	−750195	c_{37}	−21900265	c_{55}	−425262
c_2	8	c_{20}	731313	c_{38}	22204709	c_{56}	490795
c_3	−109	c_{21}	2205238	c_{39}	22429112	c_{57}	523913
c_4	−57	c_{22}	712394	c_{40}	−17328349	c_{58}	−68074
c_5	535	c_{23}	−3385572	c_{41}	−20056153	c_{59}	−261424
c_6	332	c_{24}	−3963207	c_{42}	13127166	c_{60}	−20387
c_7	−1564	c_{25}	2698262	c_{43}	16599654	c_{61}	80510
c_8	−2185	c_{26}	7140516	c_{44}	−10349321	c_{62}	14015
c_9	77	c_{27}	−2851842	c_{45}	−14388214	c_{63}	−15390
c_{10}	4426	c_{28}	−14882724	c_{46}	7141322	c_{64}	−3939
c_{11}	14632	c_{29}	−3585158	c_{47}	13005974	c_{65}	1376
c_{12}	10748	c_{30}	18011655	c_{48}	−2179835	c_{66}	706
c_{13}	−64391	c_{31}	9959742	c_{49}	−9506737	c_{67}	107
c_{14}	−114320	c_{32}	−16721296	c_{50}	−1565798	c_{68}	−98
c_{15}	120935	c_{33}	−8786675	c_{51}	4642699	c_{69}	−44
c_{16}	403948	c_{34}	23705905	c_{52}	2344706	c_{70}	12
c_{17}	72432	c_{35}	14461249	c_{53}	−1031513	c_{71}	4
c_{18}	−717244	c_{36}	−26936528	c_{54}	−1432720	c_{72}	−1

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 74$, and it is the recurrence OEIS A300608 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{72} - c_1 x^{72-1} - \cdots - c_{72}$. Its constant term is $-c_{72} = 1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 72 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 72, so $\deg q = 72$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 72, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 72, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-72 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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